

Mathematics Bootcamp

Functions, Counting, The Real Numbers

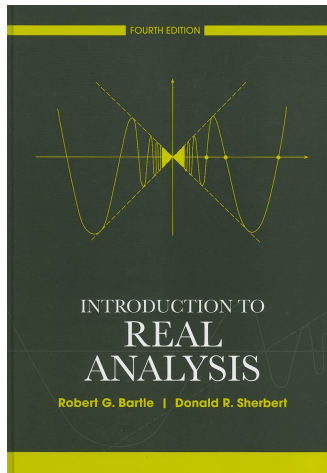
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Today

- Cartesian Products and Functions
- Finite and Infinite Sets
- The Real Numbers: Order and Completeness
- Distance, Balls, Sequences, and Compactness
- Ordinal Utility

Bartle-Sherbert - Real Analysis



- Most of what follows is taken from Bartle-Sherbert.
- My favorite introductory book for Graduate Level Math.
 - Careful attention to detail.
 - Great pedagogical order.

Cartesian Product

1.1.5 Definition

For nonempty sets A and B , the **Cartesian product** is

$$A \times B := \{(a, b) : a \in A, b \in B\}.$$

If $A := \{x : 1 \leq x \leq 2\}$ and $B := \{y : 0 \leq y \leq 1 \text{ or } 2 \leq y \leq 3\}$, the product is not a rectangle.

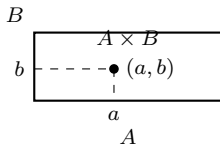


Figure 1.1.2

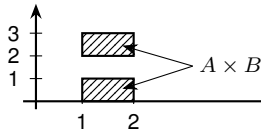


Figure 1.1.3

Functions

1.1.6 Definition

Let A and B be sets. Then a **function** from A to B is a set f of ordered pairs in $A \times B$ such that for each $a \in A$ there exists a unique $b \in B$ with $(a, b) \in f$.

$D(f) := A$ is the **domain**; $R(f)$, the set of second elements, is the **range**, with $R(f) \subseteq B$.

Functions as Graphs

- A function f maps a to *exactly one* b : $f(a) = b$ or $a \mapsto b$

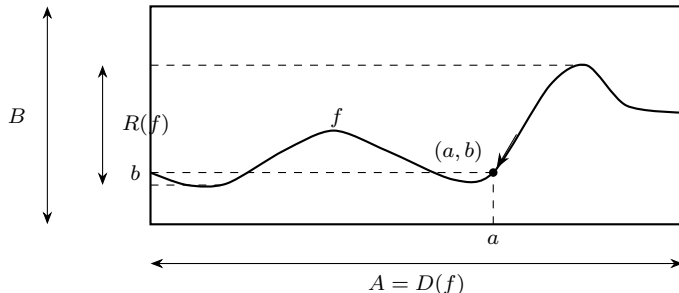


Figure 1.1.4 $A = D(f)$ A function as a graph

Injective, Surjective, and Bijective

1.1.9 Definition

Let $f : A \rightarrow B$ be a function.

- (a) f is **injective** (**one-to-one**) if $x_1 \neq x_2$ implies $f(x_1) \neq f(x_2)$.
- (b) f is **surjective** (maps A **onto** B) if $f(A) = B$; that is, if $R(f) = B$.
- (c) f is **bijective** if it is both.

Such an f is called an *injection*, a *surjection*, or a *bijection*.

Proofs - Injective and Surjective

- In order to prove that a function f is injective, we must establish that:

for all x_1, x_2 in A , if $f(x_1) = f(x_2)$, then $x_1 = x_2$.

To do this we assume that $f(x_1) = f(x_2)$ and show that $x_1 = x_2$.

- To prove that a function f is surjective, we must show that for any $b \in B$ there exists at least one $x \in A$ such that $f(x) = b$.

Composition and Inverse Images

1.1.12 Definition

If $f : A \rightarrow B$ and $g : B \rightarrow C$, the **composite function** $g \circ f : A \rightarrow C$ is defined by

$$(g \circ f)(x) := g(f(x)) \quad \text{for all } x \in A.$$

1.1.7 Definition

For $H \subseteq B$, the **inverse image** of H under f is

$$f^{-1}(H) := \{x \in A : f(x) \in H\}.$$

Cardinality - Finite and Infinite Sets

1.3.1 Definition

- (a) $\emptyset$ has 0 **elements**.
- (b) S has n **elements** if there is a bijection $\mathbb{N}_n := \{1, 2, \dots, n\} \rightarrow S$.
- (c) S is **finite** if it is empty or has n elements for some $n \in \mathbb{N}$; **infinite** otherwise.

1.3.6 Definition

- (a) S is **denumerable (countably infinite)** if there is a bijection $\mathbb{N} \rightarrow S$.
- (b) S is **countable** if it is finite or denumerable; **uncountable** otherwise.

Cardinalities of Infinite Sets

Many of our intuitions, and even formal notions, break at infinity.

Comparing Sizes

Two sets have the same cardinality when there is a bijection between them. This definition applies to both finite and infinite sets.

- For finite sets, cardinality counts the elements; for infinite sets, it grades the size of the infinity.
- $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ are countably infinite: each can be put in bijection with $\mathbb{N}$.
- $\mathbb{R}$ is uncountable: there is no bijection $\mathbb{N} \rightarrow \mathbb{R}$.

Cantor's diagonal argument proves the second claim: video.

The Role of the Real Numbers

- Economic alternatives often lie in $\mathbb{R}^L$, while utility is real-valued.
- We need the structure of $\mathbb{R}$, not a construction of it.
- Three ingredients will matter:
 - order, for comparisons;
 - completeness, for limits and extrema;
 - distance, for balls and convergence.

$\mathbb{R}$ Is a Complete Ordered Field

Ordered Field

A **field** supports addition and multiplication. A **total order** compares every pair and is compatible with those operations.

$$\begin{aligned}a < b &\implies a + c < b + c, \\ 0 < a \text{ and } 0 < b &\implies 0 < ab.\end{aligned}$$

Totality gives **trichotomy**: exactly one of $a < b$, $a = b$, $a > b$ holds.

Both $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields. Only $\mathbb{R}$ is **complete**: it has no gaps.

Bounds

2.3.1 Bounds

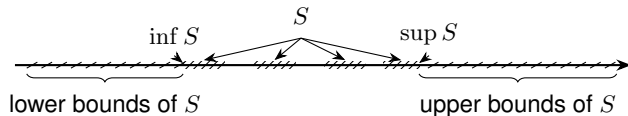
Let $\emptyset \neq S \subseteq \mathbb{R}$.

u is an upper bound $\iff s \leq u$ for every $s \in S$,

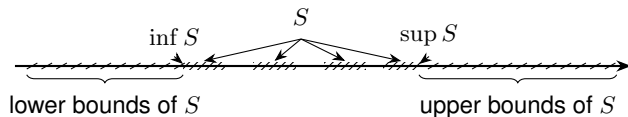
w is a lower bound $\iff w \leq s$ for every $s \in S$.

S is **bounded** if it has both an upper and a lower bound.

Example. For $S = (0, 1)$: upper bounds $= [1, \infty)$, lower bounds $= (-\infty, 0]$.



Suprema and Infima



2.3.2 Supremum = least upper bound

$$u = \sup S$$

1. $s \leq u$ for every $s \in S$.
2. $u \leq v$ for every upper bound v .

$$\begin{aligned} u \in S &\Rightarrow u = \max S; \\ u \notin S &\Rightarrow \text{no maximum.} \end{aligned}$$

Infimum = greatest lower bound

$$w = \inf S$$

1. $w \leq s$ for every $s \in S$.
2. $t \leq w$ for every lower bound t .

$$\begin{aligned} w \in S &\Rightarrow w = \min S; \\ w \notin S &\Rightarrow \text{no minimum.} \end{aligned}$$

Completeness of the Real Numbers

2.3.6 Completeness Property of $\mathbb{R}$

For every nonempty $S \subseteq \mathbb{R}$,

$$S \text{ bounded above} \implies \sup S \in \mathbb{R}.$$

Consequently, if S is bounded below, then $\inf S \in \mathbb{R}$.

Two Different Words

A *binary relation* is complete when it ranks every pair: for all x, y , $x \succeq y$ or $y \succeq x$. Nothing is left uncomparing.

The *real numbers* are complete in an unrelated sense: no bounded set is left without a supremum. Nothing is left unfilled.

Same word, unrelated properties. Days 1 and 3 use the first; Day 2 uses the second.

Key Implications for Proofs

Archimedean Property

2.4.3 *If $x \in \mathbb{R}$, then there exists $n_x \in \mathbb{N}$ such that $x \leq n_x$.*

2.4.5 Corollary *For every $\varepsilon > 0$, there exists $n \in \mathbb{N}$ such that $1/n < \varepsilon$.*

Density Results

2.4.8 Density Theorem *If x and y are any real numbers with $x < y$, then there exists a rational number $r \in \mathbb{Q}$ such that $x < r < y$.*

2.4.9 Corollary *If x and y are real numbers with $x < y$, then there exists an irrational number z such that $x < z < y$.*

Absolute Value and the Real Line

2.2.1 Definition

For $a \in \mathbb{R}$, its **absolute value**, denoted $|a|$, is

$$|a| := \begin{cases} a, & \text{if } a > 0, \\ 0, & \text{if } a = 0, \\ -a, & \text{if } a < 0. \end{cases}$$

Distance on the real line: $|a - b|$.

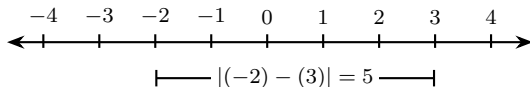


Figure 2.2.3 The distance between $a = -2$ and $b = 3$

How does a neighborhood look in two dimensions, in three, in more than three?

What Makes Something a Distance

Properties of $\|x - y\|$

The **Euclidean distance** between x and y is $\|x - y\|$, where

$$\|z\| := \sqrt{\sum_{\ell=1}^L z_{\ell}^2}.$$

For every $x, y, z \in \mathbb{R}^L$ it satisfies:

1. **Nonnegativity.** $\|x - y\| \geq 0$, and $= 0$ only when $x = y$.
2. **Symmetry.** $\|x - y\| = \|y - x\|$.
3. **Triangle inequality.** $\|x - z\| \leq \|x - y\| + \|y - z\|$.

Any function on pairs with these three properties is a **distance**. Everything that follows — balls, convergence, continuity — uses only these three.

The third is the one that does the work: no detour through y is shorter than going straight.

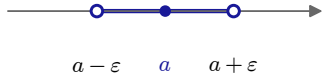
The Open Ball

Open Ball

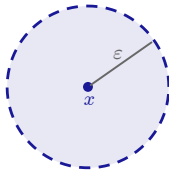
For $x \in \mathbb{R}^L$ and $\varepsilon > 0$,

$$B_\varepsilon(x) := \{y \in \mathbb{R}^L : \|y - x\| < \varepsilon\}.$$

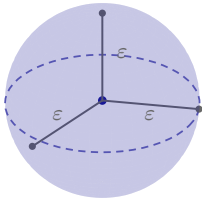
$L = 1$: an interval



$L = 2$: a disc



$L = 3$: a ball



Open, Closed, Bounded

Sets in $\mathbb{R}^L$

Let $D \subseteq \mathbb{R}^L$.

- D is **open** if every $x \in D$ has some $\varepsilon > 0$ with $B_\varepsilon(x) \subseteq D$.
- D is **closed** if its complement $\mathbb{R}^L \setminus D$ is open.
- D is **bounded** if $D \subseteq B_M(0)$ for some $M > 0$.

All three rest on the same measurement: the ball $B_\varepsilon(x)$, and so the distance $\|y - x\|$.

Sequences

Sequence

A **sequence** in $\mathbb{R}^L$ is a function $x : \mathbb{N} \rightarrow \mathbb{R}^L$. We write x_n for $x(n)$, and (x_n) or $(x_1, x_2, x_3, \dots)$ for the sequence itself.

A list of points, one for each natural number, in order. Bartle 3.1.1 is the case $L = 1$. The one question we ask of a sequence: where is it heading?

Convergence and Divergence

Convergence

(x_n) **converges** to $x \in \mathbb{R}^L$ if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N, \quad \|x_n - x\| < \varepsilon.$$

We write $x_n \rightarrow x$. A sequence that converges to no point is **divergent**.

In words: every ball around x , however small, eventually holds every term. Convergence in $\mathbb{R}^L$ is convergence in each coordinate.

Limits Keep Weak Inequalities

If $a_n \rightarrow a$ in $\mathbb{R}$ and $a_n \geq c$ for every n , then $a \geq c$.

Closed Sets Contain Their Limits

Sequential Test for Closed Sets

Let $F \subseteq \mathbb{R}^L$. Then F is closed if and only if

$$x_n \in F \text{ for every } n, \quad x_n \rightarrow x \implies x \in F.$$

A closed set cannot be left by taking limits: whatever a sequence inside it approaches is inside it too. This is how closedness is checked in practice, and how Day 3 will check it for contour sets.

Continuity of a Function

Sequential Definition

Let $X \subseteq \mathbb{R}^L$, $Y \subseteq \mathbb{R}^K$, and $f : X \rightarrow Y$. The function f is *continuous at* $x \in X$ if, for every sequence (x_n) in X ,

$$x_n \rightarrow x \implies f(x_n) \rightarrow f(x).$$

It is continuous on X if it is continuous at every $x \in X$.

Keep the objects separate. This is continuity of a *function*. Day 3 defines continuity of a *preference relation*.

Compactness in $\mathbb{R}^L$

Heine–Borel and Extreme Values

Let $D \subseteq \mathbb{R}^L$. Then

D is compact $\iff D$ is closed and bounded.

If D is nonempty and compact and $f : D \rightarrow \mathbb{R}$ is continuous, then there exist $x_{\min}, x_{\max} \in D$ such that

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}) \quad \text{for every } x \in D.$$

Back to Economics - Utility Functions

Utility Functions

In economics, we often describe preference relations by means of a *utility function*. A utility function $u(x)$ assigns a numerical value to each element in X , ranking the elements of X in accordance with the individual's preferences. This is stated more precisely in Definition 1.B.2.

Definition 1.B.2: A function $u: X \rightarrow \mathbb{R}$ is a *utility function representing preference relation* $\succsim$ if, for all $x, y \in X$,

$$x \succsim y \Leftrightarrow u(x) \geq u(y).$$

Utility Representation Implies Rationality

The ability to represent preferences by a utility function is closely linked to the assumption of rationality. In particular, we have the result shown in Proposition 1.B.2.

Proposition 1.B.2: A preference relation $\succsim$ can be represented by a utility function only if it is rational.

Is rationality enough? On finite or countable X , yes. On $\mathbb{R}_+^L$ it is not: continuity is needed too. That is Day 3.

Utility Is Ordinal

Invariance to Increasing Transformations

If u represents $\succeq$ and $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing, then $\phi \circ u$ represents $\succeq$ as well.

The numbers carry no information beyond the order they induce. Any relabelling that preserves that order represents the same preferences, so differences and ratios of utilities mean nothing.

Representation and Choice

Representation Preserves Maximizers

Suppose $u : X \rightarrow \mathbb{R}$ represents $\succeq$. For every nonempty menu $B \subseteq X$ on which a best alternative exists,

$$C_{\succeq}(B) = \{x \in B : x \succeq y \text{ for every } y \in B\} = \arg \max_{x \in B} u(x).$$

If $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing, then

$$\arg \max_{x \in B} u(x) = \arg \max_{x \in B} (\phi \circ u)(x).$$